INCH POUND

MIL-P-70951 (AR)
24 March 1992

MILITARY SPECIFICATION

PALLET, FIELD ARTILLERY PROJECTILE

This specification is approved for use, by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and quality assurance provisions for the manufacture of field artillery projectile pallets (FAPP) for use in unitizing 155MM projectiles (see 6.1).

1.2 Classification. The field artillery projectile pallet shall be classified as follows (see 6.1 and 6.2):

Type I - FAPP with latch rod assembly having rod length of 23.66 inches.
Type II - FAPP with latch rod assembly having rod length of 31.49 inches.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning Methods for Ferrous Surfaces and Pretreatments for Organic Coatings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, U.S Army ARDEC, ATTN SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8140

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MILITARY

MIL-W-12332	-	Welding Resistance, Spot, Seam and Production, for Fabricating Assemblies of Low Carbon Steel
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-A-70625	-	Automated Acceptance Inspection Equipment for Non-Electronic Components, Design, Testing and Approval of

STANDARDS

FEDERAL

FED-STD-101	-	Test Procedures for Packaging Materials

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-171	-	Finishing of Metal and Wood Surfaces
MIL-STD-1261	-	Welding Procedures for Constructional Steel
MIL-STD-1660	-	Design Criteria for Ammunition Unit Loads

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

US ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

12914619	-	Packing and Marking 155mm Projectile Pallet

(Copies of other Government documents, drawings and publications required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D3951	-	Standard Practice for Commercial Packaging

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY

ANSI/AWS B1.10	-	Guide for Non-destructive Inspection of Welds
ANSI/AWS B4.0	-	Mechanical Testing of Welds, Standard Methods for
ANSI/AWS D9.1	-	Welding of Sheet Metal, Specification for

(Application for copies of AWS publications should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P.O. Box 351040, Miami, Florida 33135.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Pallets. The pallet assemblies shall comply with all requirements specified on drawing 12914619 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Stacking strength. The pallets shall withstand a 5,000 pound minimum (min.) stacking load.

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3.4 Welding. The welding shall comply with the requirements of MIL-W-12332 or MIL-STD-1261 and weld callouts in the applicable drawings.

3.5 Weld integrity.

3.5.1 Visual inspection. The load bearing surface welds (forkwell welded to base panel) of the base assembly (see dwg. 12914620) shall be free of the following defects:

a. Undercut

b. Overlap

c. Surface cavities

d. Surface cracks in weld metal or in heat affected zone of base metal

e. Lack of weld penetration

f. Slag

g. Incomplete fusion

The requirement for porosity shall be in accordance with ANSI/AWS D9.1. The visual inspection of the welds shall be in accordance with ANSI/AWS B1.10.

3.5.2 Weld coupon bend text. The guided bend specimens shall have no open discontinuities exceeding 3.2mm (1/8 in.) measured in any direction on the convex surface of the specimen after bending, except that cracks occurring on the corners shall not be considered, unless there is definite evidence that they result from slag inclusions or other internal discontinuities.

3.6 Painted surface. The painted surfaces of the pallets shall comply with the requirements of TT-C-490, applicable drawings and with instructions specified in MIL-STD-171.

3.7 Unit load tests. The unit loaded pallets shall conform to the requirements specified in MIL-STD-1660.

3.8 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.9 Workmanship.

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3.9.1 Pallets. The pallets shall be regular, smooth, free from wrinkles, cracks, rough spots, burrs, sharp edges and any other defect that might affect the serviceability, durability and safety of the pallets. Weld splatter is permissible on non-load bearing surfaces provided it is firmly attached.

3.9.2 Paint coating. The finished coat shall cover all required surfaces and be continuous. Paint runs are permissible provided they are no more than 3 inches in length, solid and free from cracks and blisters. Excess thickness is acceptable in areas where multiple passes of the paint gun are required to obtain total coverage. Paint chips which expose primer or bare metal are acceptable provided they are not more than 0.25 inches in diameter and no more than two such paint chips on a pallet. Scratches which expose primer or bare metal are permissible provided they are no more than 1.5 inches in length and no more than 10 inches total on a pallet. Bare metal on contact surfaces of bottom of skids and in areas inaccessible to paint is permissible.

3.9.3 Parts. All parts shall be free of chips, dirt, grease, rust and foreign material. The cleaning method used shall not be injurious to any of the parts nor shall any of the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance, of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of twelve (12) painted pallet assemblies and eight (8) each of every subassembly and component.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 3	DRAWING NUMBER
	Assemblies and components		As applicable
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Latch-rod assembly (Dwg. 12914630) Examination for defects	6		4.4.2.1
	Latch-rod assembly, painted (Dwg. 12914630) Examination for defects	3		4.4.2.2
	Panel, base (Dwg. 12914623) Examination for defects	6		4.4.2.3
	Base assembly (Dwg. 12914620) Examination for defects	8		4.4.2.4
	Base assembly, painted (Dwg. 12914620) Examination for defects	3		4.4.2.5
	Panel, cover (Dwg. 12914625) Examination for defects	3		4.4.2.6
	Cover assembly (Dwg. 12914624) Examination for defects	6		4.4.2.7

NOTES

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Assemblies and components	SHEET 2 OF 3		DRAWING NUMBER As applicable NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Cover assembly, painted (Dwg. 12914624) Examination for defects	3		4.4.2.8
	Liner, cover (Dwg. 12926863) Examination for defects	6		4.4.2.9
	Spacer (Dwg. 12914629) Examination for defects	6		4.4.2.10
	Latch-handle assembly (Dwg. 12914628) Examination for defects	6		4.4.2.11
	End cap, cover panel (Dwg. 12914626) Examination for defects	6		4.4.2.12
	End cap, base panel (Dwg. 12934728) Examination for defects	6		4.4.2.13
	Trunnion (Dwg. 12934729) Examination for defects	6		4.4.2.14

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TABLE I. First article inspection MIL-P70951

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Assemblies and components	3 OF 3	As applicable
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Doubler, base (Dwg. 12914621)			
	Examination for defects	6		4.4.2.15
	Forkwell (Dwg. 12914622)			
	Examination for defects	6		4.4.2.16
	Pin, pivot (Dwg. 12914627)			
	Examination for defects	6		4.4.2.17
	Field artillery projectile pallet assembly, painted (Dwg. 12914619)			
	a) Examination for defects	8(see note 1)		4.4.2.18
	b) Stacking strength	9(see note 1)		4.5.1
	c) Unit load tests	3(see note 1)		4.5.3

NOTES

1. Samples used in their individual inspections such as painted latch rod assemblies, painted cover assemblies, painted base assemblies, cover liners, spacers and latch handle assemblies may be used to make up some of the samples which will be needed for Field Artillery Projectile Pallet assembly inspection.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of products from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials, the same material batch, and the same methods, in accordance with the same drawing, same drawing revision, same specification revision and manufactured under essentially the same conditions, and at essentially the same time. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examination and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classificati of Characteristics paragraphs. The contractor's quality program c detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

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Table II. 0-1 Attributes sampling inspection

Lot Size	Inspection Levels					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	12	4
91 to 150	*	125	32	13	12	5
151 to 280	*	125	32	30	14	6
281 to 500	*	125	32	30	17	7
501 to 1200	*	125	74	35	20	9
1201 to 3200	1250	125	74	43	24	10
3201 to 10000	1250	125	87	50	30	10
10001 to 35000	1250	296	109	61	36	10
35001 to 150000	1250	296	124	74	40	10
150001 to 500000	1250	346	156	91	40	10
500001 and over	1250	431	187	102	40	10

Numbers under inspection levels indicate sample size; asterisk (*) indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.3). When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70951 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Latch-rod Assembly			12914630
				NEXT HIGHER ASSEMBLY 12914619
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material certification for rod		3.2	4.1.2
102	Pitch diameter of thread on stud end	Level III	3.2	Gage
103	Major diameter of thread on stud end	Level III	3.2	Gage
104	Pitch diameter of thread on nut end	Level III	3.2	Gage
105	Major diameter of thread on nut end	Level III	3.2	Gage
106	Rod not threaded properly into stud	Level III	3.2	Visual-Manual
107	Outside diameter of stud bottom	Level IV	3.2	Gage
108	Thickness of stud bottom	Level IV	3.2	Gage
109	Total length of rod (Dimension A)	Level IV	3.2	Gage
110	Outside diameter of upper part of nut	Level IV	3.2	Gage
111	Outside diameter of retainer	Level IV	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.9	Visual

NOTES

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70951 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Latch-rod Assembly, Painted	1 OF 1	12914630
			NEXT HIGHER ASSEMBLY: 12914619

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Paint thickness, min.	Level III	3.2	Gage
102	Part missing or incorrectly assembled	Level III	3.2	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.9.2	Visual/Measure

NOTES.

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CLASSIFICATION OF CHARACTERISTICS

MIL-P-70951 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Panel, Base	1 OF 1	12914623
			NEXT HIGHER ASSEMBLY: 12914620

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material certification		3.2	4.1.2
102	Diameter (8 cups)	Level III	3.2	Gage
103	Position of cups (8)	Level III	3.2	Gage
104	Panel length	Level III	3.2	Gage
105	Panel width	Level III	3.2	Gage
106	Panel height	Level III	3.2	Gage
107	Position of latch-rod assembly holes (2 places)	Level III	3.2	Gage
108	Diameter of large latch-rod assembly hole (2 places)	Level IV	3.2	Gage
109	Diameter of small latch-rod assembly hole (2 places)	Level IV	3.2	Gage
110	Weld missing or incomplete	Level IV	3.2	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.9.1 3.9.3	Visual

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CLASSIFICATION OF CHARACTERISTICS

MIL-P-70951 (AR)

PARAGRAPH 4.4.2.4	TITLE Base Assembly	SHEET 1 OF 1		DRAWING NUMBER 12914620
				NEXT HIGHER ASSEMBLY 12914619
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Alignment of double hole with base panel (2 places)	Level III	3.2	Gage
102	Distance of forkwell centerline from base panel center (2 places)	Level III	3.2	Gage
103	Mating surface of weld (16 places, see Note 1).	Level III	3.5.1	Visual-Gage
104	Weld missing or incomplete	Level III	3.2	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.9.1 3.9.3	Visual

NOTES

1. This inspection pertains to welds joining the forkwell to the bottom of the base panel.

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Base Assembly, Painted	1 OF 1	12914620
			NEXT HIGHER ASSEMBLY: 12914619

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Top coat paint thickness, min.	Level III	3.2	Gage
Minor 201	Evidence of poor workmanship	Level V	3.9.2	Visual/Measure

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Panel, Cover	1 OF 1	12914625
			NEXT HIGHER ASSEMBLY 12914626

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material certification		3.2	4.1.2
102	Position of holes (8)	Level III	3.2	Gage
103	Panel length	Level III	3.2	Gage
104	Panel width	Level III	3.2	Gage
105	Panel height around edges	Level III	3.2	Gage
106	Position of outside holes in side of trough (4 holes)	Level III	3.2	Gage
107	Position of inner holes in side of trough (4 holes)	Level III	3.2	Gage
108	Width of trough (entire length)	Level III	3.2	Gage
109	Position of elongated holes (2)	Level III	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.9.1 3.9.3	Visual

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CLASSIFICATION OF CHARACTERISTICS

MIL-P-70951 (AR)

PARAGRAPH 4.4.2.7	TITLE Cover Assembly	SHEET 1 OF 1	DRAWING NUMBER 12914624
			NEXT HIGHER ASSEMBLY 12914619

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Weld missing or incomplete	Level III	3.2	Visual
102	Part missing or incorrectly assembled	Level III	3.2	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.9.1 3.9.3	Visual

NOTES

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.8	TITLE Cover Assembly, Painted	SHEET 1 OF 1	DRAWING NUMBER 12914624
			NEXT HIGHER ASSEMBLY 12914619

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Top coat paint thickness, min.	Level IV	3.2	Gage
Minor 201	Evidence of poor workmanship	Level V	3.9.2	Visual

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MIL-P-70951 (AR)

PARAGRAPH 4.4.2.9	TITLE Liner, Cover	SHEET 1 OF 1	DRAWING NUMBER 12926863
			NEXT HIGHER ASSEMBLY 12914619

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Small outside diameter (slot diameter) (Section A-A)	Level III	3.2	Gage
102	Middle outside diameter (Section A-A)	Level III	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.9	Visual

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PARAGRAPH 4.4.2.10	TITLE Spacer	SHEET 1 OF 1	DRAWING NUMBER 12914629
			NEXT HIGHER ASSEMBLY 12914619

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material certification		3.2	4.1.2
102	Radius from center line to tab (3 places) (Section A-A)	Level III	3.2	Gage
103	Location of tabs (3 places)	Level III	3.2	Gage
104	Wall thickness of sides and bottom	Level III	3.2	Gage
105	Outside diameter	Level III	3.2	Gage
106	Inside diameter of cup (Dimension B)	Level III	3.2	Gage
107	Distance from bottom surface of spacer to beginning of tab extension (3 places)	Level IV	3.2	Gage
108	Drain slots on bottom of spacer missing	Level V	3.2	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.9	Visual

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Latch-Handle Assembly	1 OF 1	12914628
			NEXT HIGHER ASSEMBLY: 12914624

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material certification		3.2	4.1.2
102	Location of detents from large hole (2 places)	Level III	3.2	Gage
103	Width at open end of handle	Level III	3.2	Gage
104	Diameter of small hole (2 places)	Level III	3.2	Gage
105	Diameter of large hole (2 places)	Level III	3.2	Gage
106	Height of embossment on large and small holes	Level III	3.2	Gage
107	Distance from center line of small hole center line of raised circular section	Level III	3.2	Gage
108	Radius of circular section	Level III	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.9	Visual

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CLASSIFICATION OF CHARACTERISTICS

MIL-P-70951 (AR)

PARAGRAPH 4.4.2.12	TITLE End Cap, Cover Panel	SHEET 1 OF 1	DRAWING NUMBER 12914626
			NEXT HIGHER ASSEMBLY 12914624

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Height	Level IV	3.2	Gage
Minor 201	Evidence of poor workmanship	Level V	3.9	Visual

NOTES

AMSMC Form 1570b-E, 1 Jul 89

Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

MIL-P-70951 (AR)

PARAGRAPH 4.4.2.13	TITLE End Cap, Base Panel	SHEET 1 OF 1	DRAWING NUMBER 112934728
			NEXT HIGHER ASSEMBLY 12914620

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Height	Level IV	3.2	Gage
Minor 201	Evidence of poor workmanship	Level V	3.9	Visual

NOTES

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70951 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.14	Trunnion	1 OF 1	12934729

NEXT HIGHER ASSEMBLY: 12914628

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material certification		3.2	4.1.2
102	Overall width at locking studs	Level III	3.2	Gage
103	Inner radius at cup	Level III	3.2	Gage
104	Outer radius at cup	Level III	3.2	Gage
105	Diameter of locking studs (2 places)	Level III	3.2	Gage
106	Length of stud (2 places)	Level III	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.9	Visual

NOTES

AMSMC Form 1570b-E, 1 Jul 89

Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.15	TITLE Doubler, Base	SHEET 1 OF 1	DRAWING NUMBER 12914621
			NEXT HIGHER ASSEMBLY 12914620

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Diameter of hole	Level IV	3.2	Gage
102	Distance from end of slot to center line of hole	Level IV	3.2	Gage
103	Width of slot	Level IV	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.9	Visual

NOTES

AM Form 1570b-E, 1 Jul 89 Repla AMSMC Form 1570, 1 Feb 85, which may not be 1.

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CLASSIFICATION OF CHARACTERISTICS

MIL-P-70951 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.16	Forkwell	1 OF 1	12914622
			NEXT HIGHER ASSEMBLY 12914620

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Overall height	Level IV	3.2	Gage
102	Width of formed box	Level IV	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.9	Visual

NOTES

AMSMC Form 1570b-E, 1 Jul 89

Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.17	TITLE Pin, Pivot	SHEET 1 OF 1	DRAWING NUMBER 12914627
			NEXT HIGHER ASSEMBLY 12914624

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Small diameter	Level IV	3.2	Gage
102	Length of small section	Level IV	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.9	Visual

NOTES

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70951 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.18	Field Artillery Projectile Pallet	1 OF 1	12914619
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Latch handle, when in closed position, does not engage (in hole) completely	Level II	3.2	Visual-Manual
102	Part missing or incorrectly assembled (see Note 1)	Level III	3.2	Visual
103	Protective coating thickness, min. (see Dwg. 12552040)	3-0-1 (Note 2)	3.6	Gage
Minor				
201	Protective finish missing incorrect or incorrectly applied	Level V	3.9.2	Visual
202	Evidence of poor workmanship (other than protective coating)	Level V	3.9.1 3.9.3	Visual

NOTES

1. Inert projectiles shall be used during inspection of assembly.
2. Three samples each shall be submitted for the cover assembly, the base assembly and the latch-rod assembly.

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4.4.3 Testing.

4.4.3.1 Stacking strength. (See 3.3), Major defect - The pallets shall be selected in accordance with the appropriate sampling plan as follows:

Lot Size	Sample Size
1-200 pallets	2
201-500 pallets	5
501-1500 pallets	8
1501-or more pallets	12

The samples shall be tested in accordance with 4.5.1. If any sample fails to meet the requirement, the lot shall be rejected. After three (3) consecutive lots have been found acceptable, every third lot shall be tested thereafter. If any of these lots are rejected, testing must be returned to three (3) consecutive lots. After three consecutive lots pass, return to testing every third lot.

4.4.3.2 Weld coupon bend test. (See 3.5.2), Major defect - During the first half of the production shift and during the last half of the production shift and whenever the electrodes are changed or dressed and when any changes are made in the control of the machine, two weld coupons shall be prepared by the welder in accordance with ANSI/AWS B4.0. If any test coupon fails to meet the requirement, when tested in accordance with 4.5.2, the quantity of pallets welded during that shift and after the earlier weld coupon test (if applicable) shall be rejected.

4.4.3.3 Primer thickness of base assembly, cover assembly and latch rod assembly. (See Dwg. 12914620, 12914624 and 12914630, as applicable), Major defect - Three test specimens (see TT-C-490), at the conclusion of each four (4) hours of primer application, shall be selected for this inspection. The primer thickness, of the specimens, shall be determined as specified in 4.5.4. If one or more specimen fails to meet the requirement, the four hour production shall be reworked and corrective action taken until production is acceptable again.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in Table I, 4.4.2.1 through 4.4.2.9 and in 4.5. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. When Automated Acceptance Inspection Equipment is to be used, the provision of MIL-A-70625 shall apply. See Section 6 of MIL-A-48078 and 6.4 herein.

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4.5 Methods of inspection (see 6.5).

4.5.1 Stacking strength. The pallet, assembled with the prototype unit load, shall be stacked with the required stacking load for a period of one (1) hour minimum. After that time, the pallet shall not have any evidence of breaking, bending, buckling or failure of any of the welds. Any pallet which fails to meet the requirement shall be classed defective.

4.5.2 Weld coupon bend test. A guided bend test, as specified in ANSI/AWS B4.0, shall be performed on the coupons. Any coupon which fails to meet the requirement shall be classed defective.

4.5.3 Unit load test. (First article inspection only) Prototype unit loads for the pallets shall be fabricated in accordance with MIL-STD-1660 with the following exceptions.

The gross weight of the ammunition load shall be 900 pounds minus 50 pounds, evenly distributed over the eight (8) cups of the base assembly and shall obtain a height as specified in Dwg. 12914619 (± 2 inches) for the assembled pallet. The following tests, on the loaded pallets, shall be performed in accordance with MIL-STD-1660 in the order specified.

4.5.3.1 Repetitive shock test. The acceptance/rejection criteria specified in section 7 of Method 5019.1 in FED-STD-101 is mandatory.

4.5.3.2 Edgewise drop (rotational) test. The acceptance/rejection criteria specified in section 7 of Method 5008.1 in FED-STD-101 is mandatory.

4.5.3.3 Incline impact test. The acceptance/rejection criteria specified in section 7 of Method 5023 in FED-STD-101 is mandatory.

4.5.3.4 Sling compatibility test. The acceptance/rejection criteria specified in section 5 of MIL-STD-1660 shall apply.

4.5.4 Primer thickness of base assembly, cover assembly and latch rod assembly. The primer thickness shall be measured by a suitable instrument.

5. PACKAGING

5.1 Preservation. None required.

5.2 Packing.

5.2.1 Commercial. The metal pallet shall be packed in accordance with ASTM-D3951.

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5.3 Marking.

5.3.1 Commercial. Shipping marking shall comply with ASTM D3951 and the following additional instructions. No shipping marking shall be placed directly on the pallets. Commercial shipping tags attached with wire shall be used unless otherwise specified. The following information shall be shown: Item name and model number, part number and revision, contract number, and date, name and address of manufacturer. If pallets are bundled, boxed or bulk-contained, the gross weight and quantity shall be included. The above information may be placed on bulk-containers or boxes by labeling or stenciling.

5.4 Shipping. When pallets from more than one lot are shipped at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing lots in transit.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This pallet is intended for use in unitizing 155MM projectiles. Latch rod assemblies of two different sizes can be used to accommodate projectiles of different sizes. The M107, M485, M804 projectiles can be unitized using one latch rod assembly (Type I pallet), while the M483A1, M718/M741, M692/M731, M825 and the XM864 can be unitized using the other latch rod assembly (Type II pallet).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.
b. Type of pallet (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. Quantity required and delivery schedules.
e. Certificates of conformance for each lot or shipment of product.
f. First Article Sample requirements.
g. Quality Conformance Inspection, if other than specified in section 4 of this specification.
h. Packaging requirements, if other than specified in section 5 of this specification.
i. Applicable stock number.
j. Serialization requirements, if applicable.

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6.3 Submission of alternative quality conformance provisions. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity responsible for the preparation of this specification.

6.4 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: SMCAR-QAR-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.5 Approval of equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: SMCAR-QAR-Q (D), Picatinny Arsenal, NJ 07806-5000. This description should include but not be limited to the accuracy and precision of the method, test data demonstrating the accuracy and precision and drawings of any special equipment required.

6.6 Subject term (key word) listing.

Ammunition
Metal
Unit load
Welding

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 8140 - A801)

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5 REASON FOR RECOMMENDATION

6 SUBMITTER

a NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7 DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME U. S. Army ARDEC Specifications And Standardization Office	b TELEPHONE (Include Area Code) (1) Commercial (201) 724-6675 (2) AUTOVON 880-6675
c ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, N. J. 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 *Previous editions are obsolete* 198/2

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE	1 DOCUMENT NUMBER MIL-P-70951	2 DOCUMENT DATE (YYMMDD) 24 MARCH 1992

3 DOCUMENT TITLE PALLET, FIELD ARTILLERY PROJECTILE

4 NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible Attach extra sheets if needed)*

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